

Customer Churn Prediction & Risk Segmentation

Business Intelligence Report | Telecom Industry

Dataset: 7,043 Customer Records | Models: Logistic Regression, Random Forest, XGBoost

1. Executive Summary

This report presents findings from an end-to-end machine learning pipeline built to predict customer churn for a telecom provider. Using a dataset of 7,043 customer records, three classification models were trained and evaluated. The best-performing model — Logistic Regression with a ROC AUC of 0.8408 — was used to segment customers into three churn risk tiers. Key findings reveal that contract type is the single most dominant driver of churn, accounting for nearly 48% of feature importance. A total of 172 customers (12.1% of the test set) were identified as High Risk, representing an immediate retention opportunity for the business.

2. Key Findings

2.1 Model Performance

Three models were trained and compared across five evaluation metrics. Logistic Regression achieved the highest ROC AUC score of 0.8408, outperforming both Random Forest (0.8346) and XGBoost (0.8235). Despite its simplicity, Logistic Regression proved to be the most reliable and interpretable model for this dataset.

Model	Accuracy	Precision	Recall	F1 Score	ROC AUC
Logistic Regression	80.4%	67.2%	54.8%	60.4%	0.8408
Random Forest	79.8%	65.9%	52.1%	58.2%	0.8346
XGBoost	78.9%	64.3%	51.5%	57.2%	0.8235

2.2 Top Churn Drivers

Based on XGBoost feature importance analysis, the following three factors were identified as the strongest predictors of customer churn:

- Contract Type (importance score: 0.486) — The most critical factor by a significant margin. Month-to-Month contract customers churn at dramatically higher rates than those on annual or two-year plans.
- Online Security (importance score: 0.073) — Customers without online security services are significantly more likely to churn, suggesting a strong link between perceived value and retention.
- Internet Service Type (importance score: 0.068) — Fiber optic users show a notably higher churn rate compared to DSL customers, possibly due to higher pricing or greater competition in that segment.

3. Customer Risk Segmentation

Using the best model's predicted churn probabilities, customers were segmented into three risk tiers. This segmentation enables the business to prioritize retention efforts based on urgency and financial impact.

Risk Tier	Customers	Avg Monthly Charges	Avg Tenure (months)	Common Contract
High Risk	172	\$79.50	14 months	Month-to-Month
Medium Risk	225	\$68.20	28 months	Month-to-Month
Low Risk	1,012	\$55.10	42 months	One/Two Year

High Risk customers represent the most urgent retention target. They are characterized by short tenure (averaging 14 months), higher monthly charges (\$79.50 average), and almost exclusively hold Month-to-Month contracts — making them easy to lose with no contractual barrier to exit.

4. Business Recommendations

Recommendation 1 — Convert High Risk Customers to Long-Term Contracts

The company should immediately launch a targeted retention campaign for the 172 High Risk customers. Offering a 15-20% discount on annual or two-year contract plans to these customers could significantly reduce churn. Since contract type is the strongest churn predictor (48% importance), even converting 30-40% of High Risk customers to longer plans would have a measurable impact on monthly recurring revenue.

Recommendation 2 — Bundle Online Security for Vulnerable Segments

Since OnlineSecurity is the second most important churn factor, the company should offer a free 3-month trial of Online Security to High Risk and Medium Risk customers who currently do not have this service. This adds perceived value to their subscription, increases switching costs, and addresses a key pain point that correlates with churn behavior.

5. Model Limitations & Future Improvements

- The dataset contains 7,043 records which, while sufficient for a baseline model, limits generalization to broader customer populations.
 - Class imbalance (approximately 73% No-Churn vs 27% Churn) may reduce recall performance. Techniques such as SMOTE or class weighting could improve minority class detection.
 - The model does not incorporate time-series behavioral data such as usage trends, support ticket history, or seasonal patterns, which could significantly improve predictive accuracy.
 - With additional data and more advanced feature engineering, ensemble stacking or neural network approaches could push ROC AUC above 0.90 in a production environment.
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